

BIOL 40C

Class 8 · The Urinary System

Reabsorption, Countercurrent, and Santiago's Diagnosis

Wednesday, April 29, 2026 · 10:00–11:50 AM · Room 8403

Dr. Giorgio Lagna · Foothill College

Bridge from Monday — Where We Left Santiago

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PATIENT CASE

Monday's setup

- Filtration membrane is broken (HUS toxin)
- GFR 25 (normal 100–120) — filtration plummets
- BUN* 110, Cr 5.2 — wastes accumulating
- Proteins + RBCs in urine (sieve damage)
- Anemia + thrombocytopenia (toxin systemic)
- BP 145/90 — paradoxically HIGH

**Blood Urea Nitrogen (BUN): measures the amount of nitrogen in the blood that comes from urea, a waste product generated by the liver during protein metabolism*

Today's questions

1. Why is his BP high?
(Hint: RAAS is firing.)
2. Would an ACE inhibitor help him?
(Hint: no, and dangerously so.)
3. Why is his urine so dark?
(Hint: ADH + countercurrent.)
4. What's the diagnosis + treatment?
(HUS → peritoneal dialysis.)

Today's Learning Objectives

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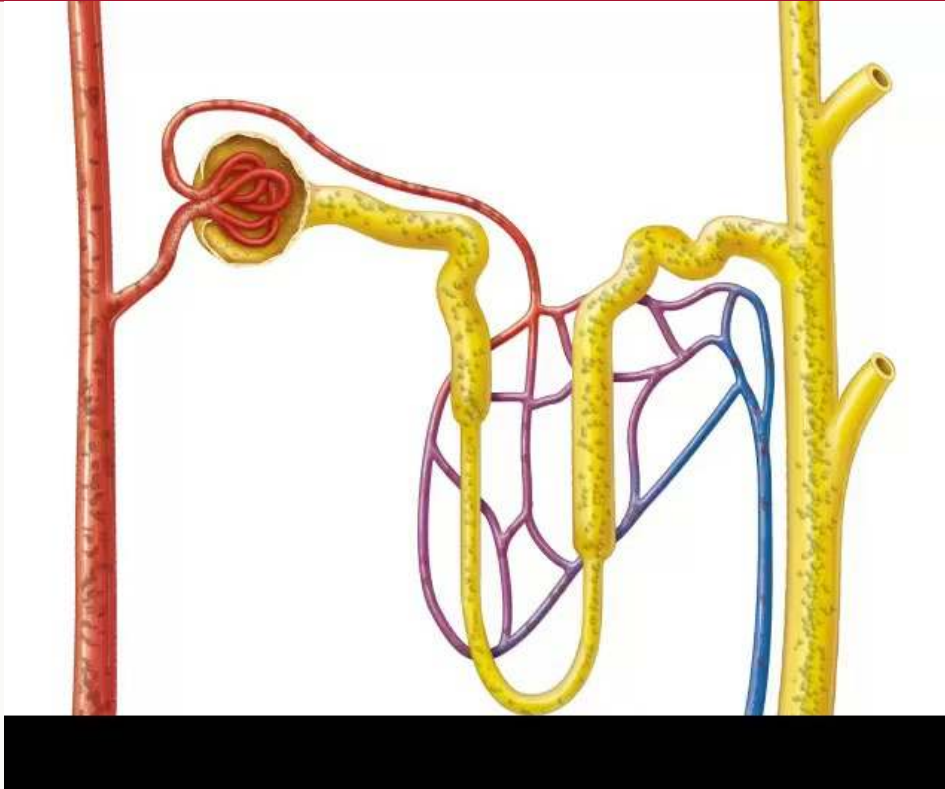
CONCEPT

- 1** Predict GFR shifts from autoregulation, RAAS, and ACE inhibitors (ALW 3a + 3b).
- 2** Explain how the basolateral Na^+/K^+ ATPase drives nearly all reabsorption.
- 3** Distinguish aldosterone, ADH, and ANP — what each saves or dumps.
- 4** Walk through the countercurrent multiplier and predict urine concentration in different ADH states.
- 5** Use creatinine to estimate GFR; explain micturition reflex and ureter anti-reflux anatomy.
- 6** Close Santiago's case: HUS → ARF → peritoneal dialysis.

Review of GFR

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CONCEPT



MODULE 24.4 (cont.)

GFR Regulation: Autoregulation + RAAS

How the kidney protects GFR — and what ACE inhibitors do

Autoregulation: Two Mechanisms

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CONCEPT

The kidney protects GFR across a wide BP range (MAP 80–180 mmHg). Two mechanisms.

Myogenic mechanism

- BP rises → afferent arteriole stretches
- Stretch triggers smooth muscle contraction
- Afferent narrows → restores GHP
- BP falls → afferent relaxes → restores GHP

Local, fast, no nervous input needed. Same physics as carotid baroreceptor reflex but executed inside the artery wall.

Tubuloglomerular feedback

- Macula densa cells (DT) sense Na^+/Cl^- flow
- High flow → paracrine signal → afferent constricts → GFR drops
- Low flow → afferent dilates AND renin released from JG cells

The macula densa is the kidney's flow sensor. It connects the distal tubule back to the corpuscle in a tight loop.

Autoregulation: Two Mechanisms

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RBF = Renal Blood Flow

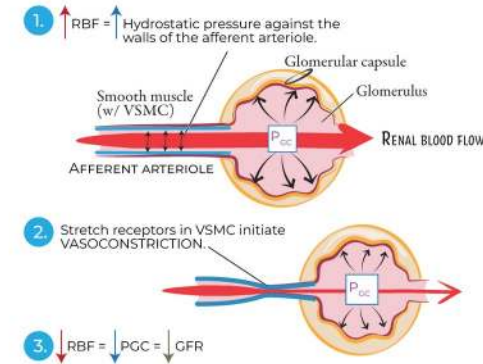
P_{GC} = GHP = Glomerular Hydrostatic Pressure

GFR = Glomerular Filtration Rate

GFR: Intrinsic Regulation

Myogenic Mechanism

Relies on inherent properties of the arterioles.

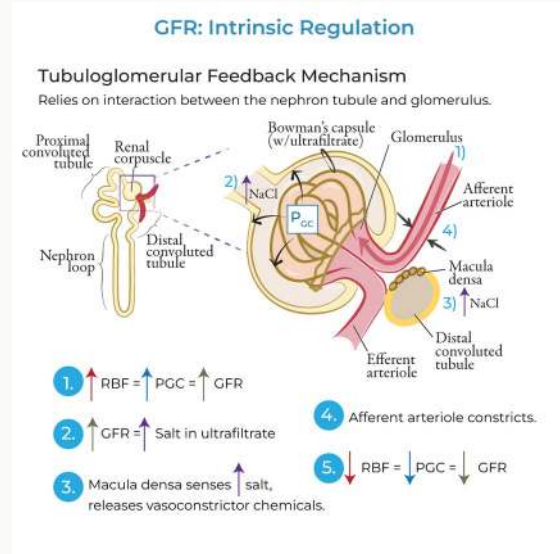


Autoregulation: Two Mechanisms

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RAAS — Renin / Angiotensin / Aldosterone

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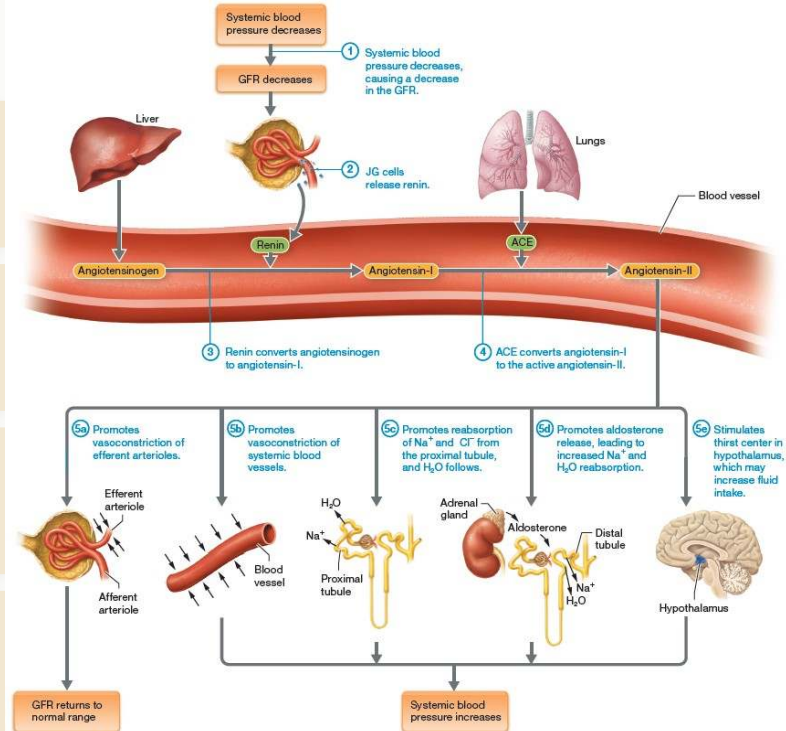
CONCEPT

1 Low BP / low Na^+ delivery JG cells release RENIN into blood.

2 Renin → Angiotensin I Renin cleaves angiotensinogen (from liver).

3 ACE → Angiotensin II ACE in lung capillaries converts AI → AII. ← ACE inhibitors block here.

4 AT-II does FOUR things Systemic vasoconstriction · efferent arteriole vasoconstriction (preserves GFR) · aldosterone release · ADH + thirst.



Three lines: Aldosterone saves salt. ADH saves water. ANP dumps both.

The Hormone Trio — Distal Tubule + Collecting Duct

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CONCEPT

Aldosterone

Source / trigger

Adrenal cortex (zona glomerulosa). Triggered by AT-II + high K^+ .

Action

Inserts more Na^+ channels + Na^+/K^+ ATPase in late DT/CD. Saves Na^+ , dumps K^+ . Water follows.

ADH (vasopressin)

Source / trigger

Posterior pituitary. Triggered by HIGH plasma osmolality + low blood volume.

Action

Inserts AQUAPORIN-2 channels in collecting duct apical membrane. Water reabsorbed osmotically. Concentrated urine, conserved water.

ANP

Source / trigger

Atrial cardiac muscle. Triggered by stretch (high BP/volume).

Action

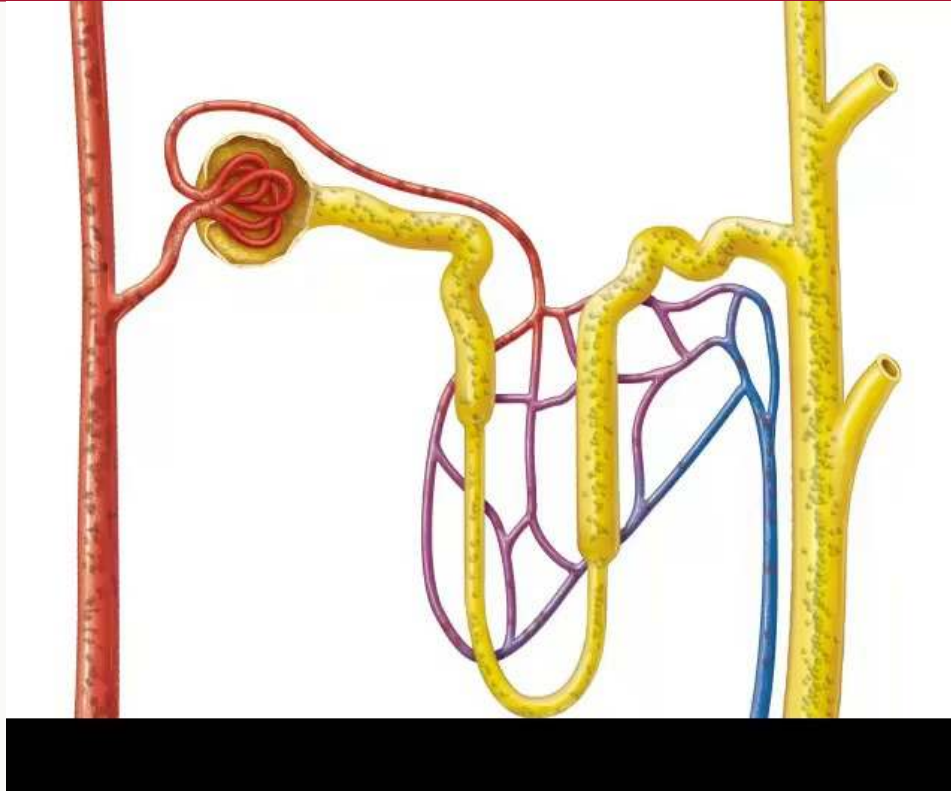
Opposes RAAS + ADH. Promotes Na^+ excretion. Diuresis. The "too-much-volume" hormone.

Aldosterone saves salt. ADH saves water. ANP dumps both.

Recap: GFR Regulation

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CONCEPT



Think-Pair-Share — ALW 3b: ACE Inhibitors

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ACTIVITY

Setup

ACE inhibitors block step 3 of RAAS (ACE in lung capillaries). Pair up. 90 seconds.



Efferent arteriole?

DILATES — no AT-II to constrict it.

GHP?

DROPS — efferent dilation lets blood out faster.

GFR?

DECREASES — less driving pressure for filtration.

Urine output?

DECREASES — less filtrate to begin with.

ALW 3b answer must use "efferent arteriole" + "glomerular filtration" — your written answer.

Santiago Beat #2 — Why an ACE Inhibitor Would Hurt Him

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PATIENT CASE

Why his BP is high

His RAAS is firing. The body senses low effective filtration → JG cells release renin → AT-II constricts arterioles + efferent → BP rises and what little GFR he has is preserved.

His hypertension is the body desperately trying to push blood through a damaged filter.

Why ACE inhibitor is wrong here

Block ACE → no AT-II → efferent dilates → GHP drops further → his already-low GFR (25) crashes further → acute renal injury worsens.

Clinical pearl: ACE inhibitors are antihypertensive AND "renoprotective" in chronic disease, but contraindicated in acute renal failure.

MODULE 24.5

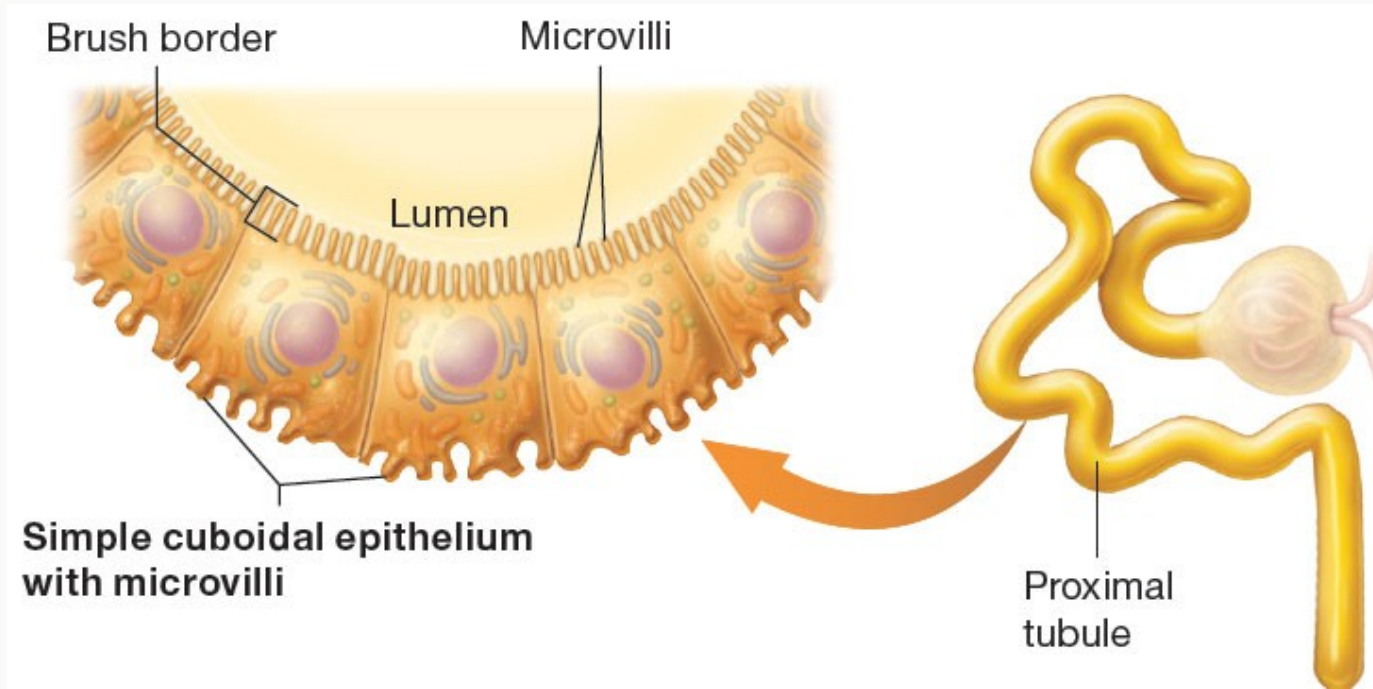
Tubular Reabsorption + Secretion

The Na^+ pump as engine + the hormone trio

What do you think the cells in the proximal tubule do?

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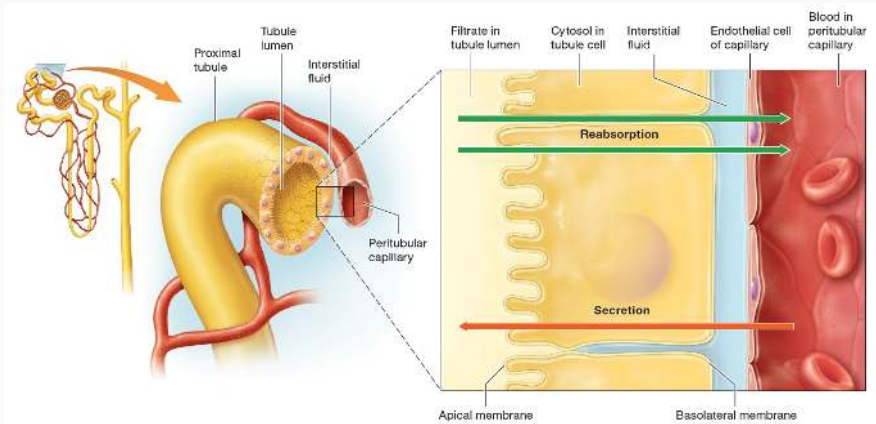
CONCEPT



Two Routes for Reabsorption

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CONCEPT



Amerman Fig 24.15 — Reabsorption + secretion principles

Transcellular

THROUGH the cell.

Apical (filtrate side) → cytoplasm → basolateral (blood side) → interstitium.

Uses transporters. Regulated.

Paracellular

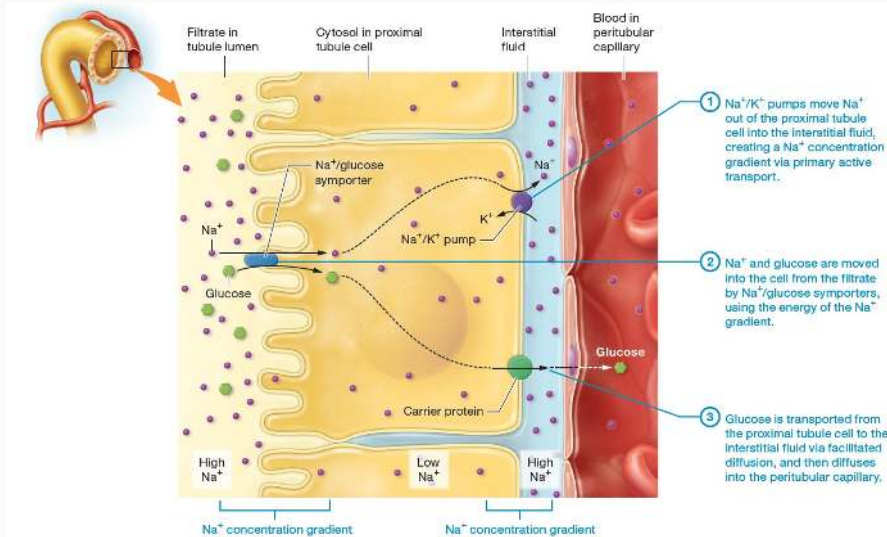
BETWEEN adjacent cells, through tight junctions.

Driven by gradients only. Less regulated. Important for water + small ions in PT.

The Na⁺/K⁺ ATPase — Engine of Reabsorption

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CONCEPT



Amerman Fig 24.16 — Proximal tubule cell

How it works

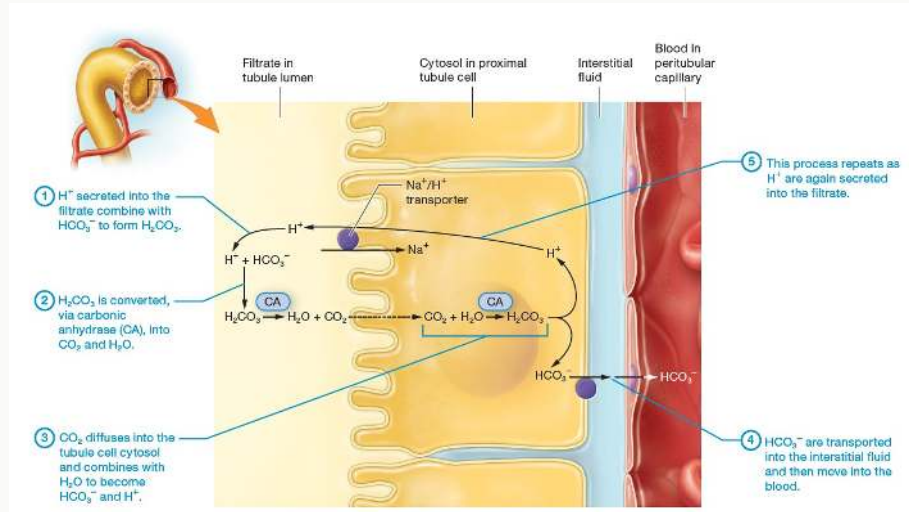
- Na⁺/K⁺ ATPase on the BASOLATERAL membrane pumps Na⁺ out of the cell into interstitium
- Keeps cytoplasmic Na⁺ LOW
- Creates a steep Na⁺ gradient across the apical membrane
- Apical co-transporters use that gradient to pull Na⁺ + GLUCOSE / AAs / phosphate INTO the cell
- Water follows osmotically — wherever Na⁺ goes, water goes

This is SECONDARY active transport — and it's the engine of nearly everything the proximal tubule does (~65% of all reabsorption).

The Na^+/H^+ antiporter — Engine of Bicarbonate Reabsorption

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CONCEPT



Amerman Fig 24.17 — Bicarbonate ion reabsorption in the proximal tubule

How it works

Bicarbonate ion reabsorption involves Na^+/H^+ antiporter activity and a reversible chemical reaction.

CO_2 , H_2O , H_2CO_3 , H^+ , and HCO_3^- react to form carbonic acid, which is then converted to CO_2 and H_2O by carbonic anhydrase on the apical tubule cell membrane.

CO_2 diffuses into the cytosol, combining with water to form bicarbonate and hydrogen ions, also catalyzed by carbonic anhydrase.

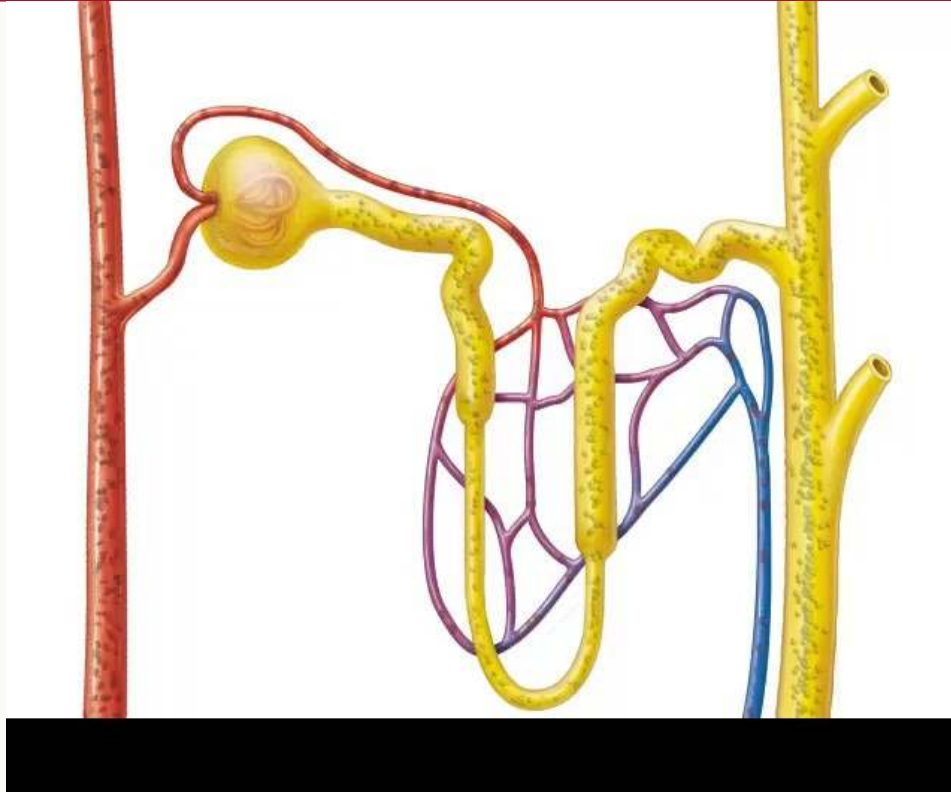
Hydrogen ions are secreted back into the tubular lumen, repeating the process.

- Proximal tubule cells reabsorb about 90% of bicarbonate ions, crucial for maintaining blood pH within 7.35-7.45.

Recap: Reabsorption and Secretion in the Proximal Tubule

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Recap: Reabsorption and Secretion in the Distal Tubule

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CONCEPT



Quick Poll — Loop Diuretics

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CHECK

Loop diuretics block the $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ co-transporter in the thick ascending limb. Predict the K^+ effect on the patient.

A. Plasma K^+ rises (hyperkalemia)

B. Plasma K^+ unchanged

C. Plasma K^+ falls (hypokalemia)

D. Depends on aldosterone status



Pollev: BIOL40C W3 C8 — Loop diuretic K^+

Answer: C — Hypokalemia

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The paradox

Blocking the co-transporter wastes Na^+ delivery to the distal tubule. High Na^+ at the distal tubule activates aldosterone-mediated Na^+/K^+ exchange. The body trades K^+ for Na^+ in the collecting duct. Plasma K^+ falls.

Clinical takeaway

Patients on loop diuretics (furosemide / Lasix) need K^+ supplementation OR a K^+ -sparing diuretic added (spironolactone, an aldosterone antagonist).

Every BP drug class — ACE inhibitors, ARBs, loop / thiazide / K^+ -sparing diuretics, β -blockers, calcium channel blockers — works on something we just covered.

MODULE 24.6

Countercurrent Mechanism

How the kidney makes urine more concentrated than your blood

The Problem the Kidney Has to Solve

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CONCEPT

To make urine MORE concentrated than blood, you need to pull water out of the filtrate without pulling solute. To pull water out by osmosis, you need a place that's saltier than the filtrate.

The trick

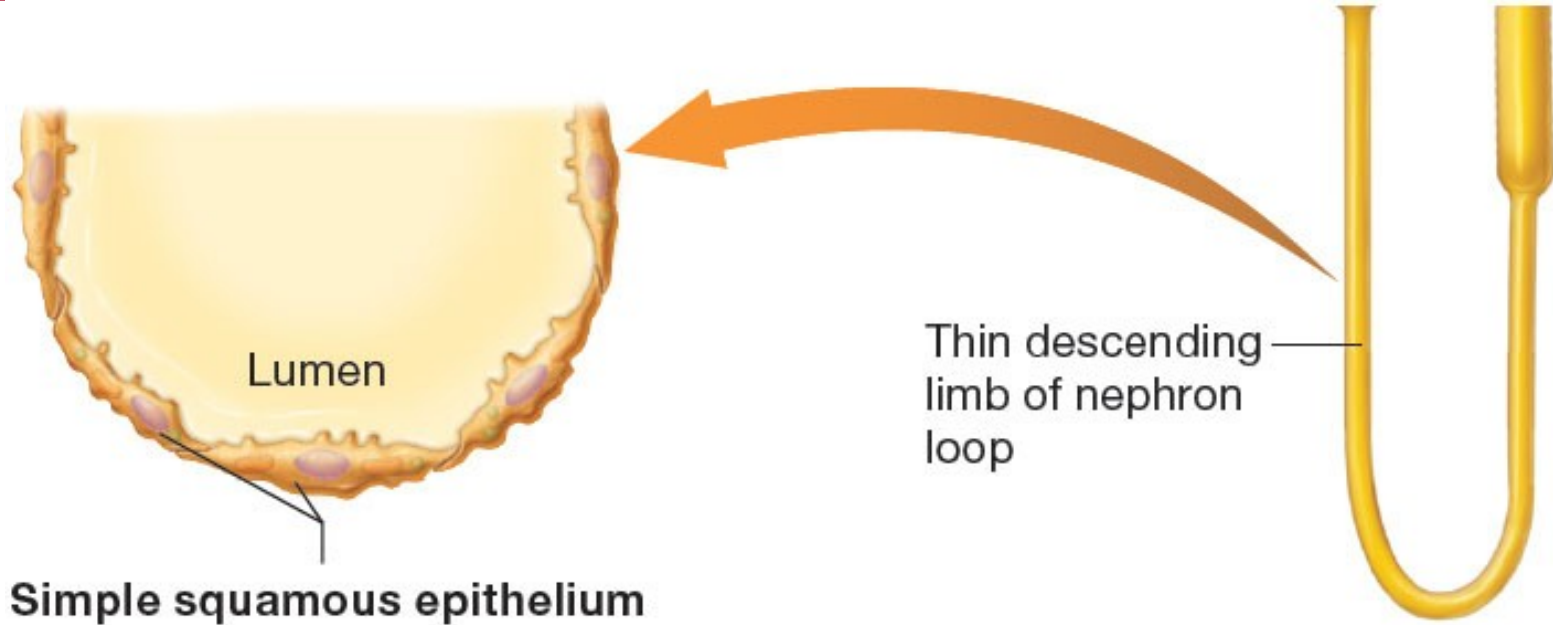
The kidney BUILDS itself a salty region — the medullary osmotic gradient. Cortex $\approx$ 300 mOsm. Deep medulla $\approx$ 1200 mOsm.

Built by the juxtamedullary nephron loops (the long ones from Class 7) using a beautifully wasteful trick: the countercurrent multiplier.

What do you think the cells in the descending loop do?

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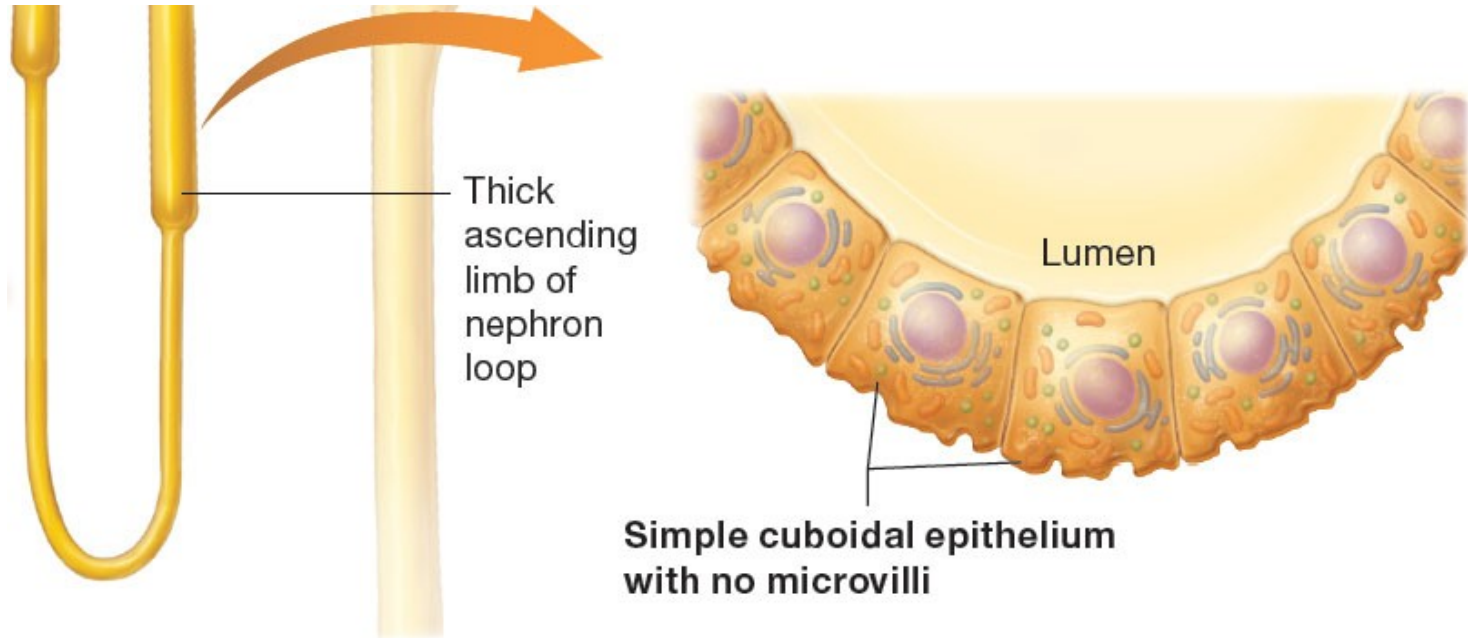
CONCEPT



What do you think the cells in the ascending loop do?

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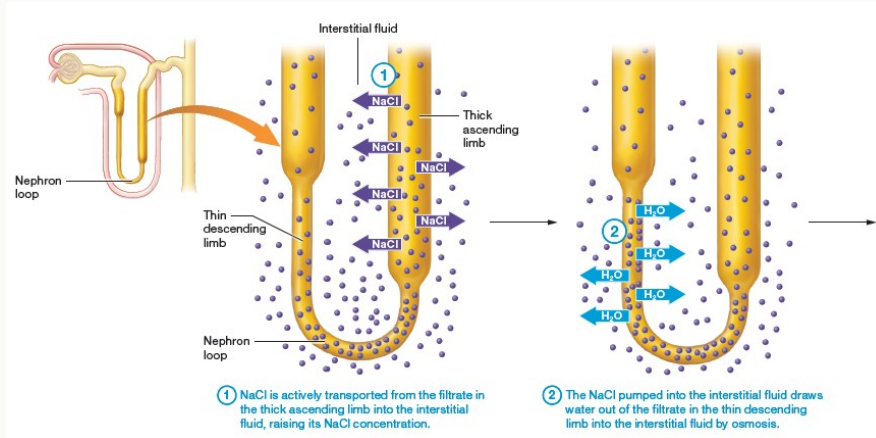
CONCEPT



Countercurrent Multiplier — Step by Step

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CONCEPT



Amerman Fig 24.21 — Countercurrent multiplier

DESCENDING limb

1

Permeable to WATER, NOT to salt. Water leaves; filtrate gets saltier as it descends.

ASCENDING limb

2

Permeable to SALT, NOT to water. Salt is actively pumped out into medulla.

Multiplier effect

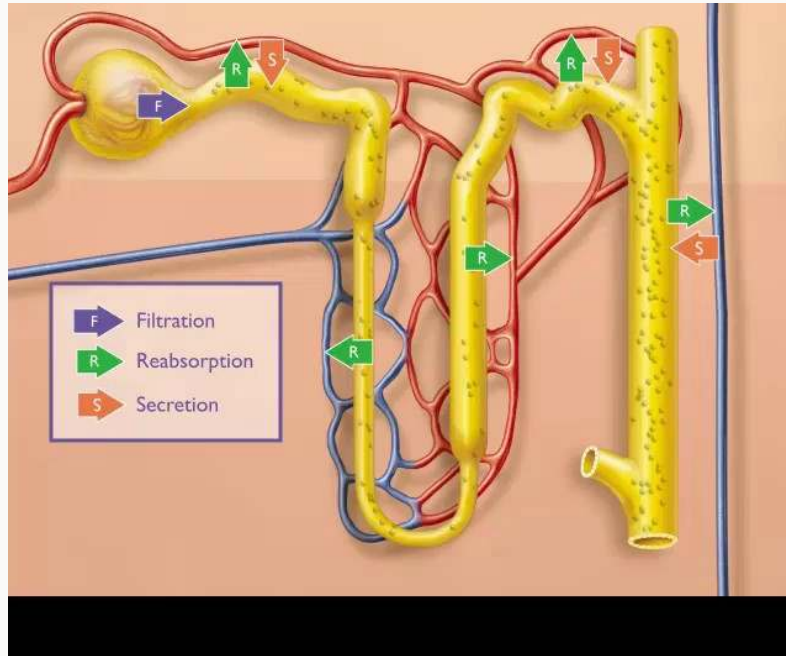
3

Salt dumped by ascending limb raises medulla osmolality, pulls more water from descending limb. Gradient self-amplifies.

Big Picture: Dilute vs. Concentrated Urine

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CONCEPT



Amerman Fig 24.23 — Production of dilute or concentrated urine

Low ADH → DILUTE urine

Aquaporins absent in CD → water can't follow gradient → large volume of dilute urine. Body says "I have too much water."

High ADH → CONCENTRATED urine

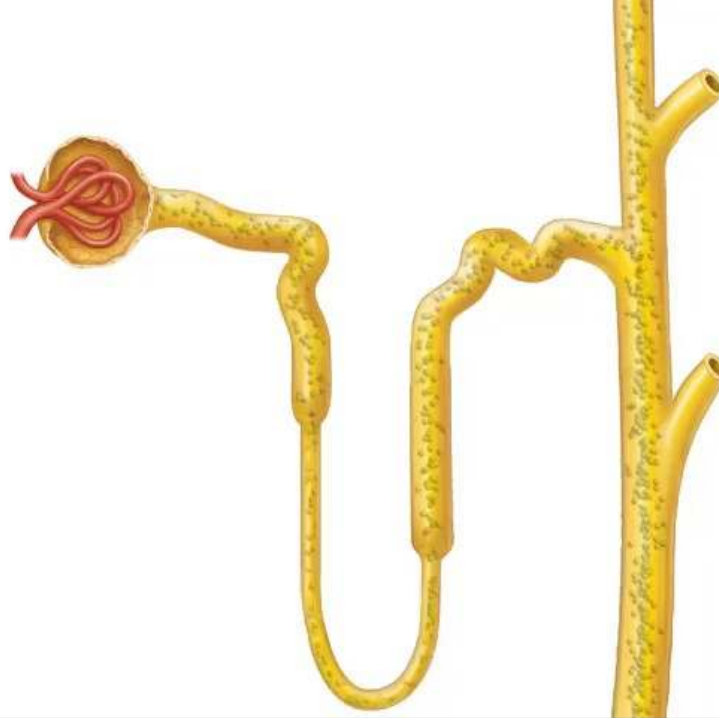
Aquaporins inserted → water sucked out by medullary gradient → small volume of dark concentrated urine. Body says "I'm dehydrated."

This is Santiago.

Big Picture: Countercurrent Multiplication and Exchange

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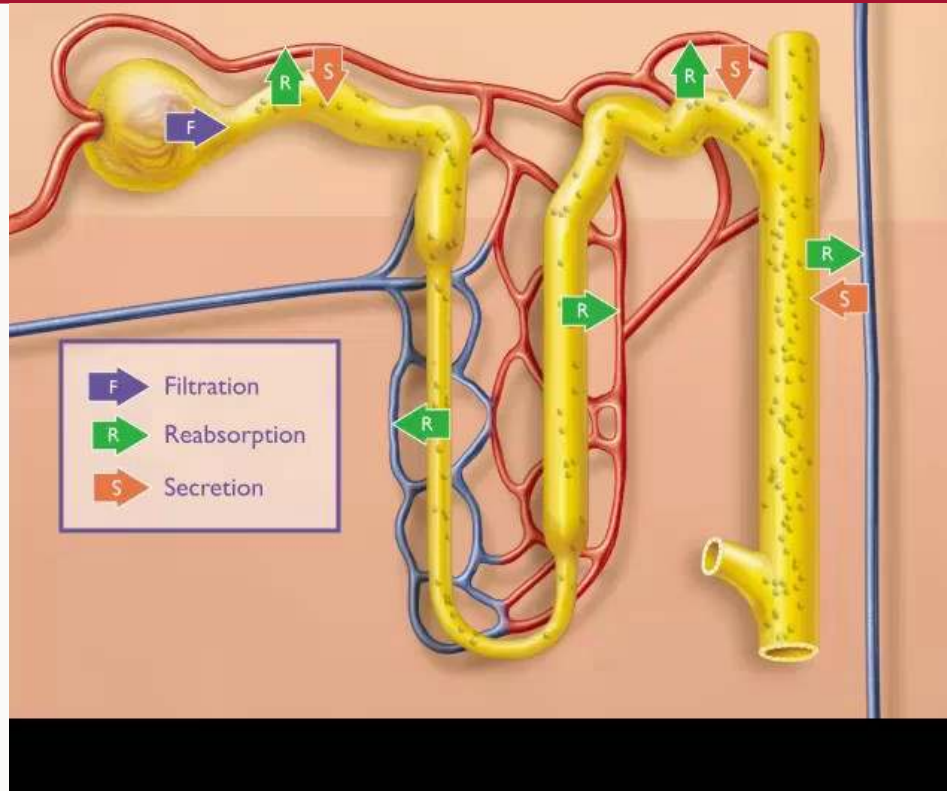
CONCEPT



Big Picture: Production of Dilute or Concentrated Urine

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CONCEPT



MODULE 24.8 + 24.9

Renal Clearance + Micturition

Why creatinine + how the bladder empties

Renal Clearance — Why Creatinine?

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CONCEPT

Renal clearance = volume of plasma fully cleared of substance per unit time • $C = (U \times V) / P$

Inulin — gold standard

- Freely filtered
- Neither reabsorbed nor secreted
- Clearance = GFR exactly

- Problem: must be infused IV. Used only in research.

Creatinine — clinical proxy

- Made at constant rate by muscle
- Freely filtered, slightly secreted
- Already in the blood — just measure plasma + urine

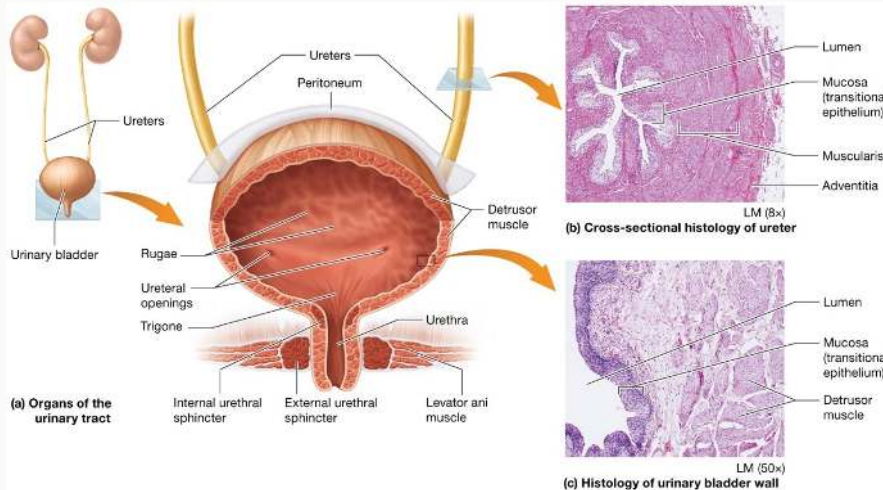
- Slightly overestimates GFR (~10%) but the trend is reliable.

- Santiago Cr 5.2 (normal 0.3–0.7) → his GFR is critically low.

Lower Urinary Tract — Anti-Reflux Anatomy

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CONCEPT



Amerman Fig 24.25 — Anatomy of urinary tract

Ureters

Peristaltic. Enter bladder at an angle. As bladder fills, the wall compresses the openings → one-way valve. Anti-reflux = protects upper tract from infection.

Bladder

Detrusor smooth muscle (3 layers). Transitional epithelium stretches. Stores ~400 mL comfortably.

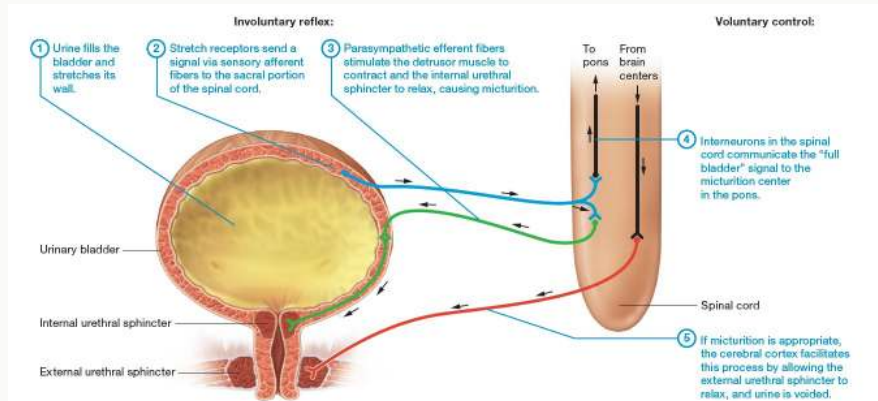
Urethra (M vs. F)

Female ~4 cm; male ~20 cm. Short female urethra → why UTIs are far more common in women.

Micturition Reflex

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CONCEPT



Amerman Fig 24.27 — Micturition

Stretch sensors fire

1

At ~200–400 mL bladder volume.

Sacral spinal reflex

2

Parasympathetic → detrusor contracts, internal sphincter relaxes.

Pontine override

3

Pontine micturition center allows VOLUNTARY release of external sphincter (somatic). This is what toilet training trains.

Closing Santiago's Case — The Whole Story

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PATIENT CASE

Zoo trip → ingestion

1

E. coli O157:H7 contamination.

Toxin enters bloodstream

2

Damages glomerular endothelial cells.

Filter broken

3

Proteinuria + hematuria. Damaged RBCs in circulation → hemolytic anemia. Platelets activated → thrombocytopenia.

GFR collapses → ARF

4

BUN, creatinine accumulate. RAAS fires → high BP despite dehydration. Distal tubule still tries: ADH on, dark concentrated low-volume urine.

Diagnosis: HUS

5

Hemolytic uremic syndrome. Stool culture confirms E. coli O157:H7.

Treatment: peritoneal dialysis

6

Peritoneum as artificial filter. Dextrose pulls fluid out by osmosis; urea + Cr diffuse out. Drain. Repeat. Kidneys heal.

Form follows function. Damage the form, the function fails. The whole organism feels it.

Today's Takeaway

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CONCEPT

Autoregulation = stability

Myogenic + tubuloglomerular feedback hold GFR steady across a wide BP range. RAAS extends the dynamic range further.

ACE inhibitors target the efferent

Lower AT-II → efferent dilates → GFR drops. Use them for chronic HTN, AVOID them in acute renal failure (Santiago).

Na⁺ pump is the engine

The basolateral Na⁺/K⁺ ATPase drives almost all reabsorption. Aldosterone tunes it; ADH tunes water.

Countercurrent makes the gradient

Long juxtamedullary loops + opposite permeabilities + parallel flow → 1200 mOsm at the medullary tip. ADH harnesses it.

Form follows function

Santiago's case is the morality tale. The kidney is built for filtration. Break the filter, break the patient.

Week 4 Preview — Fluid, Electrolyte, & Acid-Base Balance

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CONCEPT

- How the kidney + lung work together as a buffer system
- Total body water compartments + tonicity
- The math of pH — Henderson-Hasselbalch
- Metabolic vs. respiratory acidosis / alkalosis
- Compensation: fast (lung, minutes) vs. slow (kidney, hours)
- Clinical case: blood gas interpretation

Read Amerman Ch 25. New ALW + KC Quiz drops Friday — due Sunday May 3 before Class 9.

Homework — Due Saturday May 2, 11:59 PM

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CONCEPT

RCS Quiz 3

Santiago Lopez case (urinary). Reasoning, not recall.

Application Activity 3a

Stress + Kidney Filtration (Jonjei). Use NFP + autoregulation.

Application Activity 3b

ACE Inhibitors + Glomerular Filtration. Use "efferent arteriole" + "glomerular filtration".

ALW EC upload

Optional. 10 EC pts for a completed urinary workbook.

Read ahead

Amerman Ch 25 — fluid, electrolytes, acid-base. New ALW + KC Quiz drops Friday.

Lab with Dr. Schinske continues per your lab schedule. RCS readings for Wk 4 will be posted by Friday.

Class 8 — done. Week 3 — done.

"Even a damaged kidney is still trying. So are you."

Next Monday: fluid, electrolytes, and acid-base. Don't skip the reading.